

KYLA C. REAMBONANZA

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

EDUCATION

Mindanao State University–Iligan Institute of Technology (MSU-IIT) July 2026
Bachelor of Science in Computer Science – Cum Laude (CGPA: 1.65 / 1.00) *Iligan, PH*
Relevant Coursework: Artificial Intelligence, Modeling & Simulation, Parallel Computing, Data Mining, Human-Computer Interaction

PUBLICATION & RESEARCH

Simulating the Spread of AMR in Agricultural Ecosystems Oct. 2025
The 10th IEEE Cyber Science and Technology Congress (CyberSciTech) – Peer-Reviewed *Online*
• Led research and development of a parallelized agent-based simulation model using NetLogo and pyNetLogo to evaluate multi-host antimicrobial resistance (AMR) transmission dynamics. [DOI:[10.1109/CyberSciTech68397.2025.00071](https://doi.org/10.1109/CyberSciTech68397.2025.00071)]

TECHNICAL SKILLS

Programming Python, SQL, NetLogo, PHP (Laravel), JavaScript, HTML/CSS, Streamlit, Next.js
Tools & Design PyTorch/YOLOv8, Git, Bitbucket, Figma, Cloudinary, Jupyter Notebook, $\text{\LaTeX}$
Languages Cebuano (Native), Tagalog (Native), English (CEFR C1), Chinese (Novice 2)

TECHNICAL PROJECTS

Real-Time Chicken Egg Crack Detection System (Thesis) [[View Project](#)]
• Developed an automated acoustic crack detection system using Convolutional Neural Networks (CNN) in Python, converting audio signals into Log-Mel Spectrograms.
• Deployed an interactive Streamlit application featuring live inference results, confidence scores, visual spectrogram diagnostics, and automated quality logging.

Utility Pole Wire Hazard Assessment v2 (Prototype) [[View Project](#)]
• Engineered an automated risk assessment pipeline using YOLOv8 instance segmentation on urban street-level imagery, achieving ~ 0.70 Mask mAP50 on CUDA hardware.
• Implemented a clutter severity metric integrated with a real-time OpenCV Heads-Up Display (HUD).

PROFESSIONAL EXPERIENCE

Dotted Studios June 2025 – July 2025
UI/UX & Branding Intern *Iligan, PH*
• Designed user-centered digital interfaces and visual brand identities based on accessibility guidelines, wireframing, and user experience research workflows.

LEADERSHIP & COMMUNITY EXTENSION

TechForKids: Teaching Digital Literacy in Far Fetched Areas March 2026
Volunteer Facilitator *Iligan, PH*
• Collaborated with a team of volunteer educators to navigate logistical challenges and foster inclusive, encouraging learning environments for local underserved youth.

Computer Science Society June 2025 – May 2026
Undersecretary, Publication & Documentations Committee *MSU-IIT*
• Facilitated instructional support and assisted lead educators in delivering technical concepts to student officers, promoting digital inclusion and technical documentation.

College of Computer Studies Community Extension Program March 2025 – June 2025
Volunteer Facilitator *Iligan, PH*
• Instructed male Persons Deprived of Liberty (PDLs) on fundamental programming logic and computational problem solving using Scratch.